

Ketamine as an Antidepressant: Does Stereospecific Synthesis Matter?

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A Brief History

Ketamine (nicknamed “Special K”) is a derivative of a class of compounds called arylcyclohexylamines. These compounds are psychoactive and are most notable for their strong dissociative effects.¹ Development of the arylcyclohexylamine class began with the synthesis of phencyclidine (PCP) in 1956, where it showed promise as an anesthetic. However, strong psychological side effects disqualified it from being clinically viable. Ketamine was developed in 1962 in hopes of retaining PCP’s abilities as a CNS depressant without the harsh side effects.²

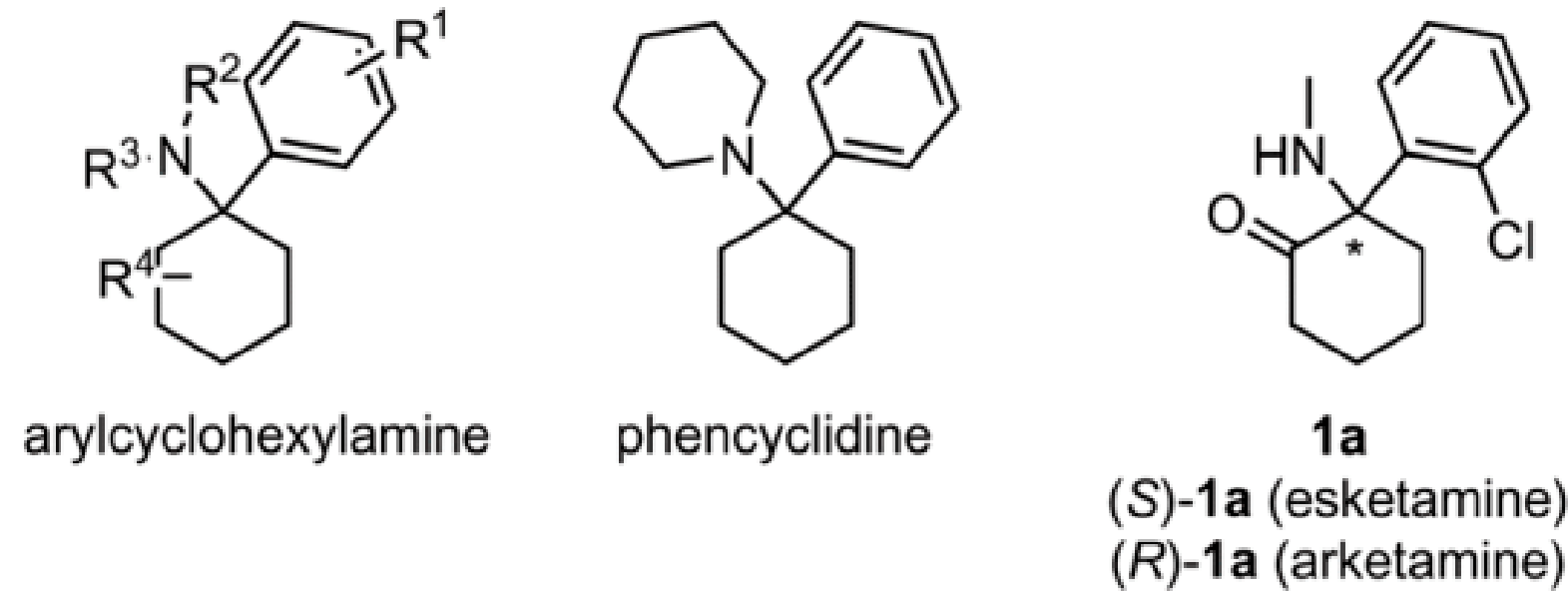


Figure 1. Arylcyclohexylamine family structures.³

In the early 2000s, ketamine began to be explored within psychiatry as a potential antidepressant, particularly for treatment-resistant depression (TRD).¹ Ketamine is a chiral molecule and has historically been administered as a racemic mixture containing equal parts R- and S- enantiomers, also called arketamine and esketamine, respectively. Only within the last decade has research begun to distinguish the unique pharmacological profiles of the two enantiomers.⁴

The most recent update came in 2019, when Johnson and Johnson received FDA approval for an esketamine nasal spray called Spravato[®], which remains the only FDA-approved ketamine intervention that can be used in a psychiatric context. Racemic ketamine continues to be utilized as an anesthetic but has also begun being used off label to treat TRD.⁶ However, ongoing research suggests enantiomerically pure forms and their downstream metabolites may offer distinct therapeutic advantages. Consequently, racemic ketamine and its individual enantiomers are treated as separate pharmaceutical entities, which has necessitated the development of distinct synthesis and manufacturing processes.^{4;7}

Mechanism of Action

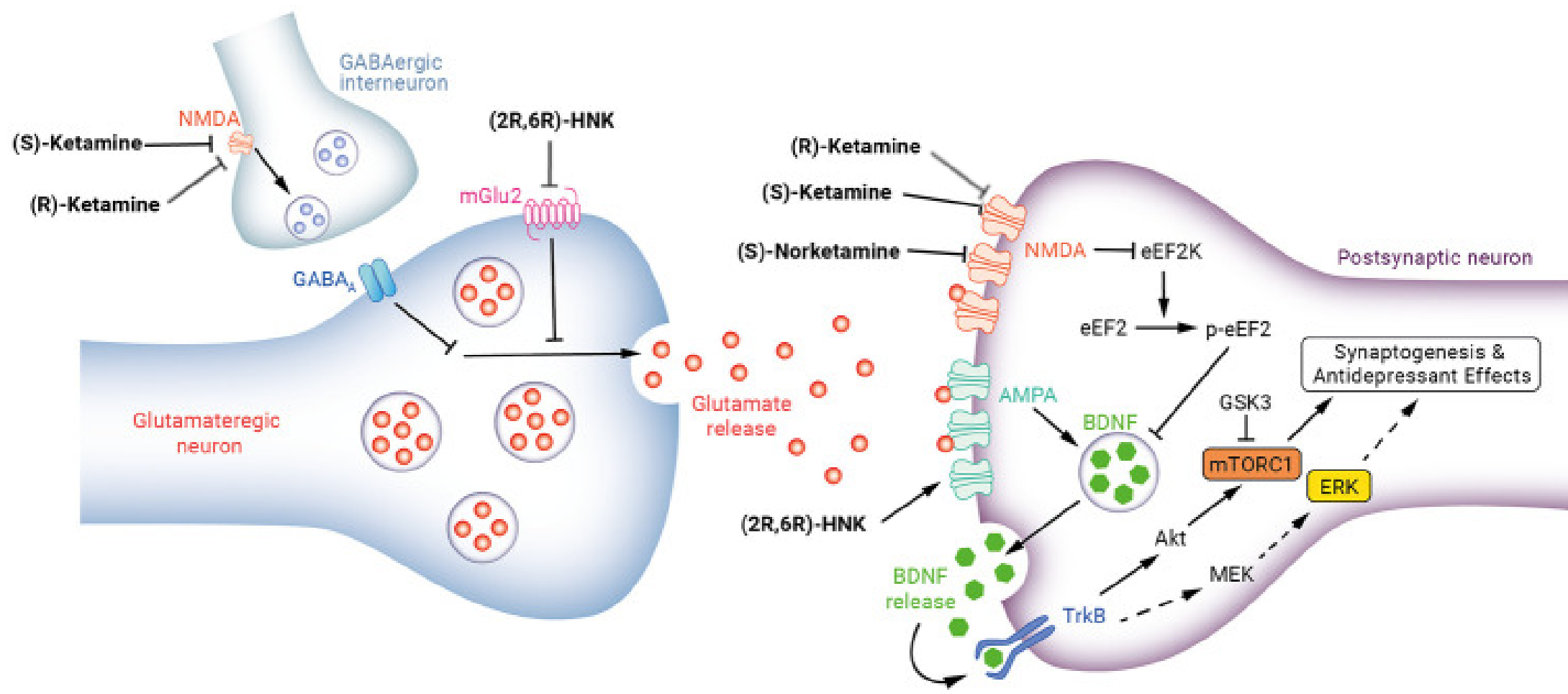


Figure 2. Ketamine’s main mechanim of action. It’s main role is as an uncompetitive NMDA inhibitor. This inhibits the release of the inhibitory neurotransmitter GABA, resulting in increased glutamate release (also known as a glutamate surge). Glutamate then binds AMPA receptors on the post-synaptic neuron, leading to a signalling event that results in increased neural connections being formed (synaptogenesis). This particular effect is responsible for ketamine’s antidepressant effects, as new neural connections can be formed in order to erode old harmful habits. Additionally, enantiomeric forms and their downstream metabolites have also been reported to add to these effects via inhibition of pre-synaptic metabotropic glutamate (mGlu2) autoreceptors and direct inhibition and activation of post-synaptic NMDA and AMPA receptors, respectively.⁴ Schematic adapted from⁴

Synthesis

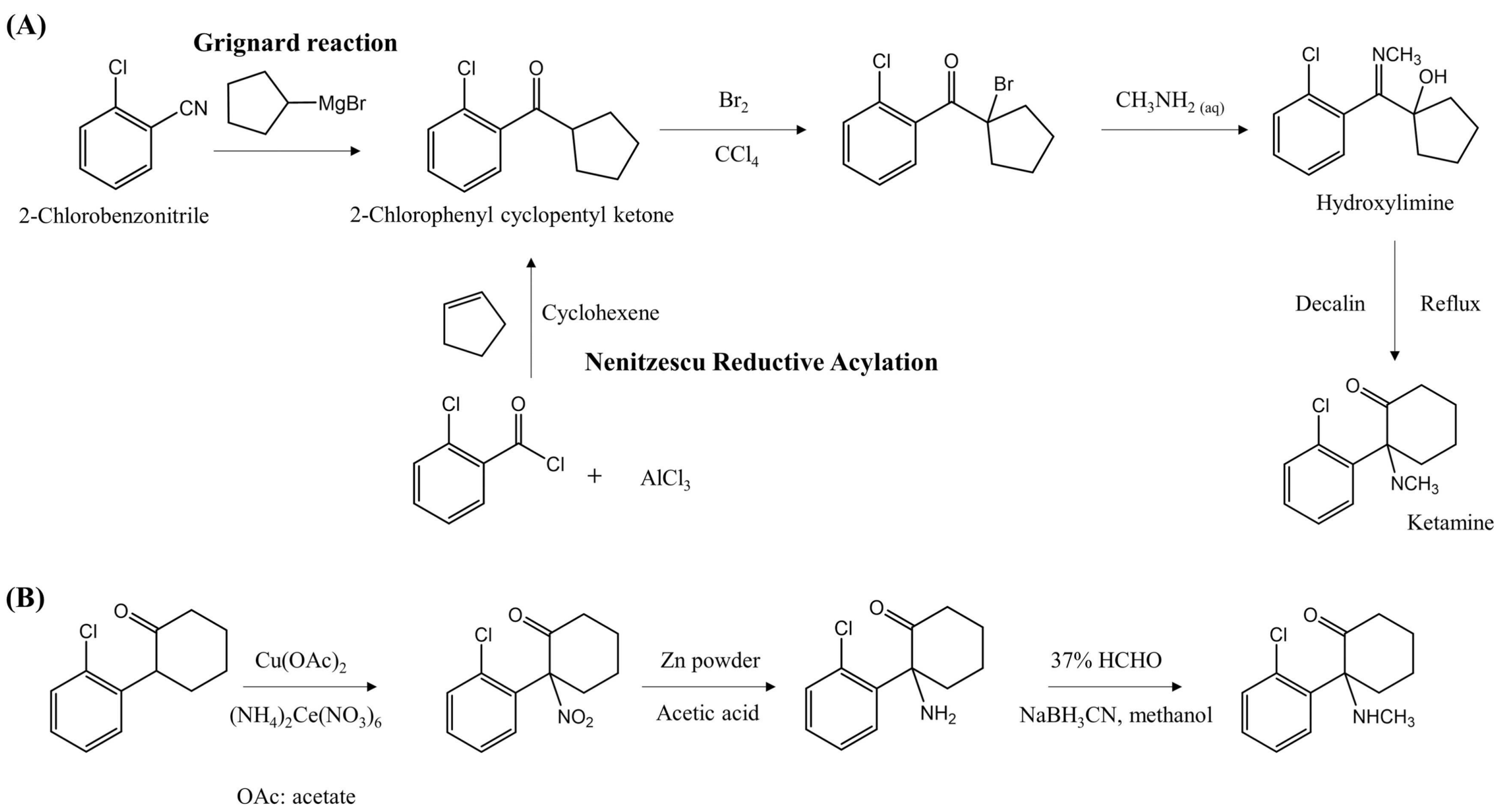


Figure 3. (A) Foundational arylcyclohexylamine-based synthetic pathway first reported in the 1960s, forming the basis of early pharmaceutical racemic ketamine manufacture.⁸ In 2020, the grignard reaction was substituted with nenitzescu reductive acylation to improve ease of synthesis, increase yield and drastically decrease price.⁹ (B) Alternate synthesis pathway for racemic ketamine developed in 2017. Reflects contemporary pharmaceutical manufacturing priorities; including efficiency, safety, and regulatory compliance.¹⁰ Schematic reproduced from¹¹

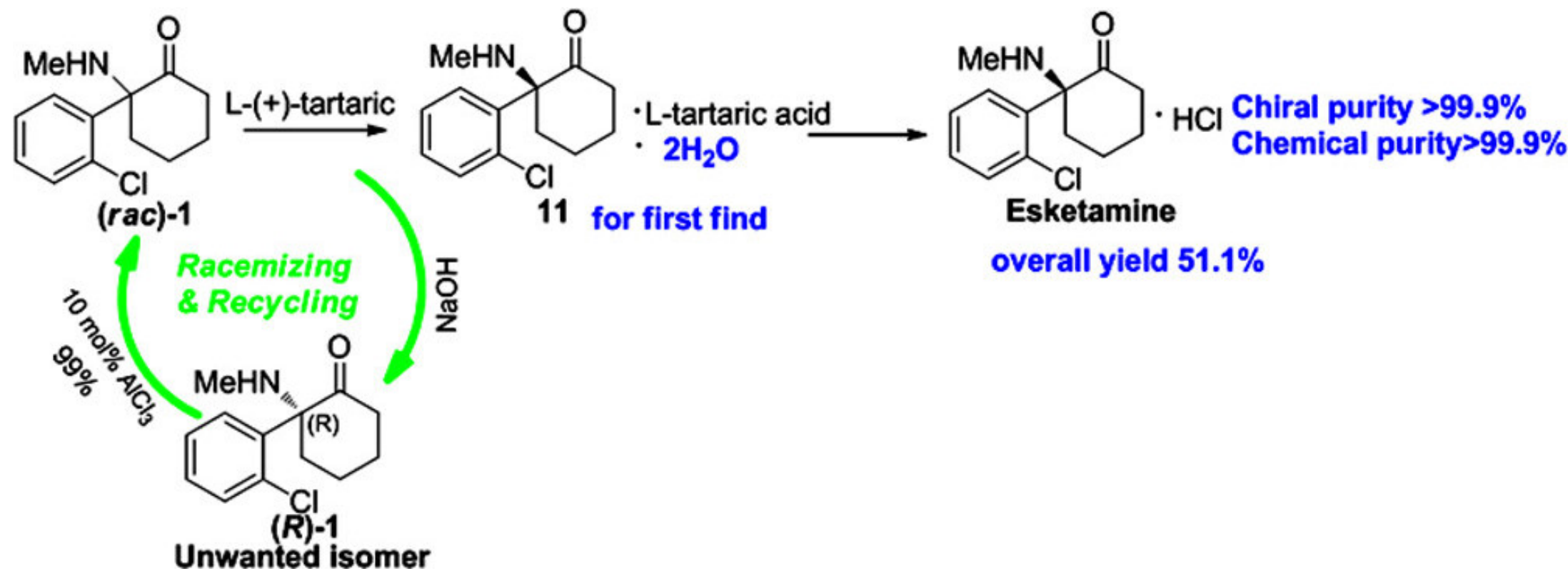


Figure 4. Industrial resolution-racemization strategy for the production of esketamine, developed in 2020 following the clinical approval of intranasal (S)-ketamine.⁹ Schematic adapted from⁹

Metabolism

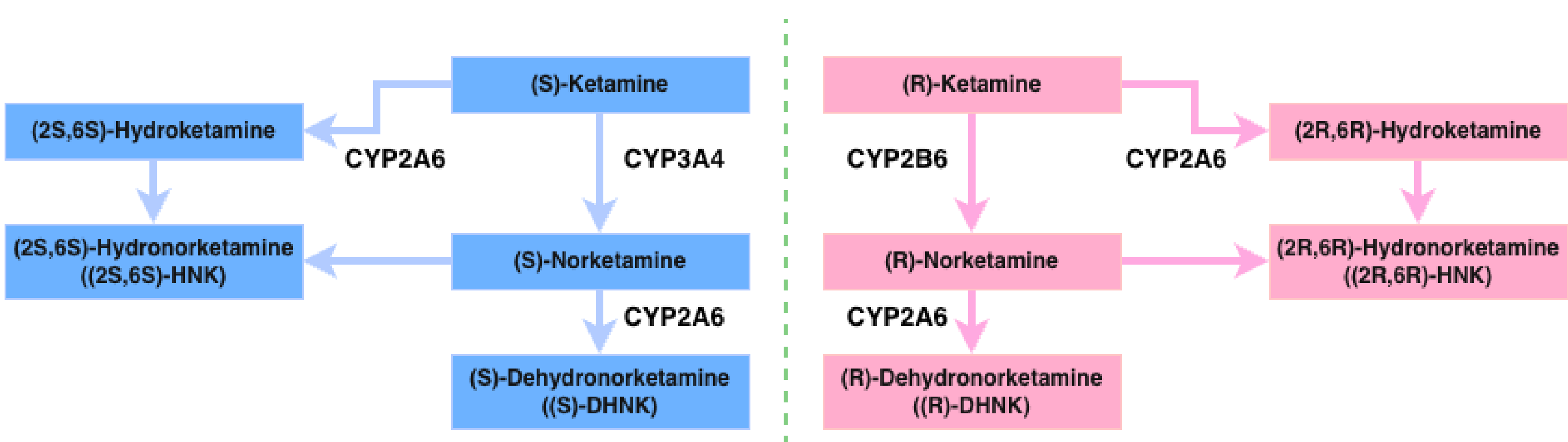


Figure 5. Both ketamine enantiomers undergo rapid and extensive metabolism by cytochrome P450 enzymes in the liver. This diagram shows the major ketamine metabolites and corresponding liver enzymes that have been reported to contribute to the pharmalogical activity profile of ketamine.^{4;12} Schematic adapted from⁴

Efficacy

Ketamine’s primary antidepressant utility lies in its rapid onset, a critical property for patients experiencing acute suicidal ideation.¹³ Consensus regarding ketamine’s role in long-term maintenance of depression remains unclear. Most research suggests that its effects are primarily acute, with limited evidence for sustained benefit beyond 2–3 weeks.^{14;15} Some meta-analyses indicate that repeated intranasal administration may support longer-lasting antidepressant effects, likely due to improved bioavailability and patient adherence compared with intravenous dosing.¹³ Conversely, other studies have reported that intravenous delivery can be more effective, as it allows for weight-adjusted dosing (see Table 2).¹⁵

Table 1. NMDA receptor affinities of ketamine enantiomers.¹³

Compound	NMDA Receptor Affinity (μM)
Esketamine	0.3
Arketamine	1.40

More broadly, while ketamine’s rapid antidepressant effects can be life-saving in acute situations, long-term management generally requires integration with additional interventions, such as psychotherapy or conventional antidepressants.¹³ The potential differences in efficacy between administration methods also complicate the evaluation of whether racemic ketamine or its enantiomerically pure forms are superior. Esketamine, with its higher NMDA receptor affinity, enables effective intranasal delivery at lower doses (see Table 1).⁴ Early research suggests that arketamine may primarily drive the antidepressant response, whereas esketamine may contribute more to dissociative side effects. This uncertainty regarding which enantiomer or downstream metabolite is responsible for specific pharmacological effects highlights the need for further studies to confirm these observations.⁴

Regulatory Capture as a Driver of Drug Price Inflation

Direct price comparisons between generic ketamine and esketamine (Spravato[®]) are challenging, as generic ketamine costs reflect drug acquisition only and exclude administration, supervision, and facility fees, whereas esketamine is delivered as a commercially available, self-administered intranasal formulation. Nevertheless, the substantial disparity in cost per effective dose highlights the influence of differential regulatory and reimbursement pathways on pricing.¹⁶

Table 2. Comparison of per-dose costs for generic ketamine and esketamine (Spravato) in treatment-resistant major depressive disorder.¹⁷

Drug	Effective Dose	Acquisition Cost (per dose)	Administration Method
Ketamine (generic) ^a	0.5 mg/kg	\$3.64–\$5.75	Intravenous infusion
Esketamine (Spravato [®])	56–84 mg	\$546–\$819	Intranasal spray

^a Acquisition costs assume a 70 kg adult (35 mg dose). Cost ranges reflect variation in purchasing prices and vial sizes, including partial vial use and bulk purchasing.

Implications and Future Directions

Ketamine demonstrates clear potential as a rapid-acting antidepressant, particularly in acute and treatment-resistant contexts. However, its precise role in long-term depression management remains unresolved, with ongoing uncertainty regarding optimal enantiomeric composition, delivery method, and the contribution of downstream metabolites. Clarifying these factors and aligning regulatory frameworks accordingly will be essential to reducing barriers to access and enabling cost-effective, evidence-based use.